

MADHAV MOHAN

AI Engineer | GenAI Engineer | LLM Engineer | AI Platform Engineer

Chennai, India | madhavam60@gmail.com | +91 9952028821 | [linkedin.com/in/mohan60](https://www.linkedin.com/in/mohan60) | github.com/maddy333

PROFESSIONAL SUMMARY

AI/GenAI/LLM Platform Engineer (MSc, UCD) with 3.5 yrs shipping production GenAI systems: sub-15ms LLM inference for 12M+ users, \$3K/mo cost elimination via self-hosted vLLM, and multi-agent RAG orchestration across LangChain, LangGraph, Kubernetes, and Terraform. Netherlands HSM-visa eligible — immediate relocation.

TECHNICAL SKILLS

LLM & GenAI: LangChain, LangGraph, Agentic AI, Multi-agent orchestration, RAG (Hybrid & Semantic), LLM Fine-Tuning (QLoRA, LoRA, PEFT), Quantization (NF4), Prompt Engineering, Function Calling, Embedding Models, vLLM, Ollama, OpenAI API, Anthropic API, Transformers

LLMOps & MLOps: MLflow, Model Registry, Evaluation & Benchmarking, Drift Monitoring, CI/CD (GitHub Actions, GitLab), Inference Optimization, CUDA, RAGAS, LLM Guardrails, PyTorch, TensorFlow, HuggingFace, Scikit-learn, bitsandbytes, Asyncio

Retrieval & Vector Store: Qdrant, OpenSearch, Pinecone, Milvus, FAISS, ChromaDB, BM25 Re-ranking, Reciprocal Rank Fusion, Embedding Pipelines

Platform & Infrastructure as Code: AWS (EKS, EC2, S3, IAM, VPC), Terraform, Kubernetes (K8s, KEDA, HPA), Helm, Docker, Ansible, Kafka, Temporal.io, Redis, Prometheus, Grafana, OpenTelemetry, FastAPI, REST API, Python, SQL, Azure, Bash

EXPERIENCE

AI Platform Engineer | SportsMechanics India | Chennai | Oct 2025 – Present

- Built the company's first IaC practice (Terraform, Helm, Ansible) for the AWS platform (EC2, IAM, IGW, S3) — cutting environment provisioning from 5 hrs to under 2 min and enabling reproducible dev/test/prod deployments.
- Built an NLP semantic video search engine (LLMs + RAG + FAISS) over a 20-year match corpus, replacing a 93-parameter manual filter workflow and reaching 96% retrieval accuracy.
- Built a real-time AI commentary pipeline on quantized local models (EC2 g4dn, Ollama/vLLM, CUDA), eliminating external LLM API costs and saving \$3,000/month.
- Built a real-time player-performance dashboard for 12M+ active users at sub-15ms P99 latency via optimized vLLM inference on Kubernetes with KEDA autoscaling and Prometheus/Grafana SLOs.
- Delivered real-time win-probability adjustments within 50ms per delivery using lightweight transformer models on high-velocity stadium radar streams.

GenAI Engineer | ShortCastle Technologies | Remote | Jan 2024 – Sep 2025

- Built a production agentic AI system (LangChain + dense retrieval + tool/function calling) over 500K+ chess positions — 60% improvement in automated query resolution.
- Built an end-to-end MLOps platform (Docker, GitHub Actions CI/CD, model versioning, zero-downtime rollouts, Prometheus/Grafana alerting) sustaining 99.9% uptime over 18 months.
- Built an "AI Chess Coach" chatbot using RAG over global chess literature for contextual, natural-language post-game analysis.
- Eliminated 100% of third-party API costs via a self-hosted local LLM pipeline delivering real-time chess strategy analysis.

AI Research Intern | Microsoft Academic-Industry Project, UCD | Dublin, Ireland | Jun 2023 – Aug 2023

- Built a fraud-detection ML system (BiLSTM + BERT) achieving 98% accuracy on the Welfake benchmark — deployed to Azure via FastAPI + Docker with CI/CD automation.
- Built NLP feature pipelines fusing sentiment signals (VADER, TextBlob, Afinn) with DistilRoBERTa dense emotion embeddings.

Associate Data Engineer | Guardian Life Insurance | Chennai | May 2021 – Jul 2022

- Built Azure Data Factory + Databricks ETL pipelines processing 200 GB/day — 80% throughput improvement and 45% reduction in Spark SQL query runtime.
- Automated analyst reporting workflows (Alteryx), eliminating 65 hrs/week of manual effort.

PROJECTS

Distributed RAG Pipeline — Event-Driven Ingestion with KEDA Autoscaling github.com/maddy333/kafka-keda-rag-pipeline

- Event-driven RAG ingestion platform (Kafka → KEDA-autoscaled K8s workers → Qdrant) with semantic chunking and dead-letter-queue fault tolerance, deployed via Terraform + Helm — the open-source blueprint behind the 12M-user, sub-15ms production pipeline. *Stack: Python, FastAPI, Kafka, Kubernetes, KEDA, Qdrant, Terraform, Helm.*

GenAI Gateway — Multi-Region Active-Active LLM Routing github.com/maddy333/gateway

- Enterprise LLM gateway with SHA-256 semantic caching (\$0 cost on repeat queries), per-tenant FinOps budget enforcement, and circuit-breaker failover between self-hosted vLLM and OpenAI — the blueprint behind the \$3K/month cost-elimination result. *Stack: Python, FastAPI, Redis, HTTPX, OpenTelemetry, Prometheus.*

QLoRA Fine-Tuning for Edge Inference github.com/maddy333/fencingllm

- Fine-tuned Llama via QLoRA (4-bit NF4 quantization) for edge deployment — JSON schema compliance 65%→98%, VRAM 5.8GB→1.6GB, throughput 22→45 tok/s; experiments tracked in MLflow. *Stack: QLoRA, LoRA, PEFT, CUDA, MLflow, Transformers.*

Agentic AI — Autonomous Multi-Agent Control Plane (HITL) github.com/maddy333/agentic_ai

- Multi-agent control plane on Temporal.io for durable, crash-resilient execution with LangGraph state-machine agents, async human-in-the-loop safety overrides, and immutable PostgreSQL audit logging. *Stack: Python, Temporal.io, LangGraph, FastAPI, SQLAlchemy, PostgreSQL.*

EDUCATION

- MSc Computer Science — University College Dublin, Ireland | 2022 – 2023
- B.Tech Information Technology — SRM University, Chennai | 2016 – 2020

CERTIFICATIONS

AWS Certified Data Engineer – Associate | Microsoft Certified: Azure Data Engineer Associate (DP-203)